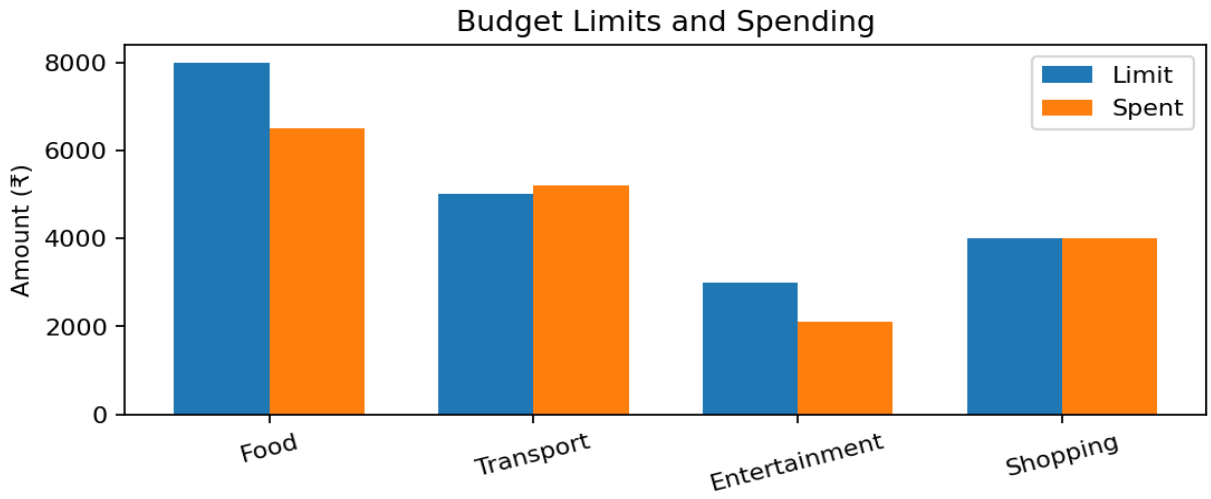
4.3 Smart Budget Planning

Budget Limit Setting

Budget limit setting allows users to establish a maximum amount that they want to spend in each category. The assistant continuously compares recorded spending with the selected limit. If spending reaches or exceeds the limit, the system can display an alert. This helps users notice overspending early and adjust their spending.

Objectives

- Create spending limits for categories.
- Record or enter current spending.
- Compare spending with limits.
- Generate alerts when limits are reached.
- Help users control unnecessary expenses.



Working Procedure

1. Create categories such as food and transport.
2. Set a monthly limit for each category.
3. Enter current spending.
4. Compare current spending with the limit.
5. Display an alert when the limit is reached or exceeded.

Example Data Table

Category	Monthly Limit	Spent	Status
Food	■8,000	■6,500	Within limit
Transport	■5,000	■5,200	Limit exceeded

Category	Monthly Limit	Spent	Status
Entertainment	■3,000	■2,100	Within limit
Shopping	■4,000	■4,000	Limit reached

Python Implementation

```
limits = {
"Food": 8000,
"Transport": 5000,
"Entertainment": 3000,
"Shopping": 4000
}

spent = {
"Food": 6500,
"Transport": 5200,
"Entertainment": 2100,
"Shopping": 4000
}

for category in limits:
amount = spent[category]
limit = limits[category]

if amount >= limit:
print(category, ": ALERT - limit reached/exceeded")
else:
remaining = limit - amount
print(category, ": Remaining =", remaining)
```

Benefits

Spending limits provide an early warning system. Alerts can help users recognize overspending before the end of the month and make informed adjustments.

Note: All amounts are illustrative examples for academic demonstration. A real application can collect data through forms, a database, or transaction records.